Quarantyne Academy

Cybersecurity Homework – Week 1

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Course: Risk Management Framework / Security Foundations

WEDNESDAY HOMEWORK – March 4, 2026

Multiple Choice

1. **Which device acts like a traffic controller for internet data in your home network**?

A. Router
B. Keyboard
C. Printer
D. Monitor

Answer: A

2. **Which of the following translates website names into IP addresses?**

A. DNS
B. HTTP
C. VPN
D. FTP

Answer: A

3. **Which of the following shows a secure website connection**?

A. http://
B. https://
C. ftp://
D. tcp://

Answer: B

4. **Which layer model explains how data travels across networks?**

A. TCP Model
B. OSI Model

C. Server Model
D. Network Model

Answer: B

Short Research Question

5. **Research one famous cyber attack that affected a company or government. Explain what happened, who was affected, and what the impact was. Minimum: 150 words.**

The 2021 Colonial Pipeline ransomware attack was one of the most serious cyber attacks against U.S. critical infrastructure. A hacker group called DarkSide gained access to Colonial Pipeline's network using a compromised VPN password. The account did not have multi-factor authentication, which made it easier for the attackers to enter the system. Once inside, they deployed ransomware that forced the company to shut down its pipeline operations to contain the attack.

The shutdown affected millions of people across the East Coast of the United States. Colonial Pipeline supplies about 45% of the region's fuel, so gas stations, airlines, and businesses that rely on fuel deliveries were impacted. Several states began experiencing fuel shortages within days.

The impact was significant. Gas prices increased, panic buying started at gas stations, and the U.S. government declared a regional emergency. Colonial Pipeline paid about $4.4 million in bitcoin to restore operations, though investigators later recovered part of the payment.

LAB – Phishing Email Analysis

Find two phishing emails online (or examples from research). Identify suspicious links, grammar errors, and fake sender addresses. Explain how you knew they were phishing emails.

Phishing Email Example 1

Description
A phishing email pretends to come from a company's IT department. The email tells employees that their password will expire and asks them to reset it immediately using a link.

Suspicious Indicators

Fake Sender Address
The email appears to come from IT but the sender address is different from the company domain.

Example:
it-support@company-security.com

Suspicious Link
The email includes a link to reset the password. When checked, the link goes to a different website instead of the official company login page.

Grammar Errors
The message contains poor wording such as:
"Your password will expired today. Click link now to keep access."

Why It Was Phishing
The email tries to trick users into entering their login credentials on a fake website. If the user enters their information, the attacker can steal the credentials and access the company's systems.

Phishing Email Example 2

Description
A phishing email pretends to come from a bank and tells the user that their account has suspicious activity. The message asks the user to click a link and verify their account information.

Suspicious Indicators

Fake Sender Address
The email address looks similar to a bank but uses a public domain.

Example:
security-bank@gmail.com

Suspicious Link
The email includes a link that directs users to a fake website designed to collect login credentials.

Example:
www.bank-secure-login.com

Grammar Errors
The email may contain poor wording such as:
"Your account has been locked due suspicious activity."

Why It Was Phishing
The email attempts to scare the user into clicking a link and entering sensitive information. Attackers use this method to steal usernames, passwords, or financial data.

PROJECT – OWASP Attack Scenario

Create a realistic attack scenario involving SQL Injection or Cross-Site Scripting (XSS). Explain how the attacker performs the attack, what the attacker gains, and how developers can prevent the attack. Minimum: 250 words.

SQL Injection

A small online shopping website has a login page where customers enter their username and password to access their accounts. The website stores user information in a database. However, the website does not properly check or validate what users type into the login fields. This creates a security weakness that attackers can exploit.

An attacker visits the login page and enters malicious code instead of normal login information. For example, the attacker types:
' OR '1'='1
into the username or password field. Because the website does not validate the input, the database interprets the code as part of the login request. The condition 1=1 is always true, so the database allows access without checking the real password. As a result, the attacker may gain unauthorized access to user accounts or even administrative accounts.

Once inside the system, the attacker can access sensitive information stored in the database. This may include customer names, addresses, emails, and payment information. In more serious cases, the attacker could modify or delete records, download the entire database, or create new administrator accounts to maintain long-term access.

Developers can prevent SQL injection by validating all user input and using parameterized queries or prepared statements. These methods treat user input as data rather than executable commands. Developers should also apply the principle of least privilege, so the database account used by the application only has limited permissions.